Appx_25_On_Selection_Law_STUB_v1.0 (Trailhead)

Title: The Selection Law — Survivorship Across Domains
Status: STUB (Trailhead; synthesis node; required for AoC coherence)

Goal

Synthesize a single Selection Law that explains why certain invariants (π, c, α, mass ratios, etc.) persist as Survivors across recursion, and provide a universal decision framework for classifying any proposed constant as:

- [SURVIVOR] (selected invariant),
- [EFFECTIVE] (regime-dependent bookkeeping),
- [TUNED] (inserted/fit),
- [UNRESOLVED] (insufficient evidence).

This appendix is the bridge between AoC_00 (operator + Units Firewall + discriminator posture) and the full Atlas (AoC_1…AoC_n).

Core Claim (provisional)

Constants are not "inputs." They are attractors of survivable structure: outcomes of repeated collapse/extension/selection cycles under $\zeta/\omega/\xi$. A constant persists when it is:

1. Stable under admissible perturbations,
2. Minimal under description/encoding pressure, and
3. Cross-regime consistent (does not require contradictory redefinitions across domains).

Must Include

A) Cross-domain selection consistency

Define a common evaluation rubric that can be applied to:

- π (closure/eigenmode of geometric coherence),
- c (null-slope mapping boundary / causal limit),
- α (dimensionless coupling survivor),
- mass ratios (proton/neutron; survivable scaling outputs),
- gravity coupling (α_G or equivalent).

Deliverable: one-page Selection Consistency Matrix (constants × criteria × pass/fail/unknown).

B) Universal discriminator protocol (template)

A standard “HoR-ready” recipe for any AoC entry:

- Claim: what invariant is being asserted (dimensionless form required).
- Units Firewall: what is allowed to vary vs must remain invariant.
- Prediction family: what must be true if it’s a Survivor (not just a fit).
- Discriminator test: measurement/analysis recipe + control for look-elsewhere.
- Decision rule: explicit kill condition(s).
- Promotion gate: when does it move from STUB → Appendix → AoC canon?

Deliverable: a reusable Discriminator Sheet template for the Atlas.

C) Meta-level falsifiers (framework killers)

List conditions that would falsify selection-as-explanation itself, e.g.:

- No nontrivial overlap window exists where π/c/α/mass ratios can all be Survivors under one consistent rule-set.

- Constants require mutually incompatible survivorship criteria (selection becomes ad hoc).

- Discriminator protocols systematically fail: “Survivors” are only post-hoc fits.

- A competing model predicts the same outcomes with fewer assumptions (AoC loses parsimony).

Deliverable: Selection Law Kill-Switch List (ranked by severity + testability).

Outputs

- Selection Consistency Matrix (AoC-wide rubric)
- Universal Discriminator Sheet (HoR packaging template)
- Meta-Falsifiers / Kill-Switch List
- Cross-links: “where each AoC entry plugs in” (dependency mini-map)

Primary Dependencies

- AoC_00_Atlas_Preface (epistemic tags + promotion discipline)
- AoC_0_The_Collapse_Operator (selection mechanism + falsifier posture)
- Appx_10_On_Extension_Bounds_STUB (survivability window overlaps)

- Appx_17_On_Renormalization_and_EFT_STUB (effective vs invariant clarity)
- Core AoC entries: AoC_1 (π), AoC_2 (c), AoC_3 (α), AoC_4–5 (mass ratios), AoC_6 (gravity)

Support

Appx_25_SelectionLaw_Support/
(selection matrices, discriminator templates, cross-domain audits, counterexample catalog, rewrite targets, HoR packaging plan)

Collapse Consequence

If selection cannot be stated as a single, cross-domain survivorship law with universal discriminators and real kill-conditions, then constants revert to tuning, coherence becomes narrative, and persistence (of intelligences, memories, and universes) loses its explanatory spine.